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Numerov-Padé scheme for the one-way Helmholtz equation in tropospheric radio-wave propagation

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***Abstract*—This paper is devoted to applying the higher-order finite-difference approximations to the tropospheric radio-wave propagation problem. It is proposed to use the 4th order approximation (Numerov discretization) of the transverse differential operator and rational Padé approximations of the propagation operator. A discrete transparent non-local boundary condition that can take into account the modified refractive index of the troposphere has been constructed. An error analysis of the considered scheme with emphasis on tropospheric propagation has been carried out. A comparison with the existing finite-difference schemes and the split-step Fourier method has been performed. The effectiveness of the proposed approach has been shown.**

***Index Terms*—Padé approximation, Parabolic equation, transparent boundary, Numerov discretization, radio-wave propagation**

I. Introduction

ONE of the most widely used approaches to solving the tropospheric radio-wave propagation problem is the parabolic equation (PE) method [1], [2], [3]. The PE method allows simultaneously taking into account spatial variations in the tropospheric refractive index, terrain profiles, and very arbitrary parameters of the transmitting antenna. This method is continually being modified to account for more and more complex propagation conditions [4]. It becomes possible to account for backscattering [5], [6], consider propagation through woodlands [7] and over the rough sea surface [8]. There are works devoted to applying the PE method in three-dimensional problems and in the time domain [9]. It follows that the corresponding numerical methods for the PE must also be continuously improved.

There are two commonly used methods for solving the PE: split-step Fourier (SSF) method [10], [11] and finite-difference (FD) approximations [12], [13]. The SSF method is the most widely used, but recently more and more works have been devoted to applying FD methods for solving the PE in tropospheric propagation problems [14], [15], [8], [16]. The SSF method is considered faster but has difficulties with modelling both the upper and lower boundary conditions (BC). FD methods are better for handling complex BC. Detailed comparative analysis of these methods has been carried out in [11], [17], [18], [19].

A distinctive feature of the considered problem is the unbounded computational domain. For modeling the upper transparent BC within the FD methods, transparent nonlocal BC (NLBC) is used [20], [21], [22], [23]. SSF method handles the upper transparent BC using an artificial absorbing layer, which leads to an incorrect solution in some cases [16], [24]. Modeling the lower impedance BC also may be an issue within the SSF method [25], [26]. Implementing impedance BC within the FD methods is straightforward [2].

In this paper, we use a combination of FD Padé approximations [27], [24] by the longitudinal coordinate (x) and 4th order Numerov approximation [28] by the transversal coordinate (z). This work aims to improve the efficiency of the FD approximations of the PE.

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II. Higher-order PE method

The evolutionary (step-by-step) solution of the one-way Helmholtz equation is written using the propagation operator as follows [2]

$$u(x+\Delta x,z)=\exp\left(ik\Delta x\left(\sqrt{1+L}-1\right)\right)u(x,z), \quad (1)$$

$$Lu=\frac{1}{k^2}\frac{\partial^2 u}{\partial z^2}+\left(m^2-1\right)u,$$

where $u(x,z)$ is the desired field component, $m^2(x,z)$ is the modified refractive index of the troposphere, $k=2\pi/\lambda$ is the wavenumber, λ is the wavelength.

The following notation is further used

$$u_j^n=u(n\Delta x,j\Delta z),\ u^n(z)=u(n\Delta x,z).$$

We use an arbitrary order Padé approximation $[m/n]$ of propagation operator (1) [27], [29]

$$\exp\left(ik\Delta x\left(\sqrt{1+L}-1\right)\right)u\approx \frac{1+\prod_{l=1}^{m}\tilde{a}_l L^l}{1+\prod_{l=1}^{n}\tilde{b}_l L^l}u=\prod_{l=1}^{p}\frac{1+a_l L}{1+b_l L}u, \quad (2)$$

where a_l and b_l are approximation coefficients, which are determined using special numerical approaches [29]. In order to calculate these coefficients high-precision arithmetic has been used.

By introducing p-1 auxiliary functions $v_1^n(z),v_2^n(z),\ldots,v_{p-1}^n(z)$, the action of propagation operator (1) may be rewritten as a system of similar one-dimensional differential equations that are solved sequentially

$$\begin{cases}\left(1+b_1L\right)v_1^n=\left(1+a_1L\right)u^{n-1}\\ \left(1+b_lL\right)v_l^n=\left(1+a_lL\right)v_{l-1}^n \qquad l=2,\ldots,p-1.\\ \left(1+b_pL\right)u^n=\left(1+a_pL\right)v_{p-1}^n\end{cases} \quad (3)$$

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The 4th order Numerov scheme [30], [28] approximates the second derivative as follows

$$\frac{\partial^2}{\partial z^2} \approx \frac{1}{\Delta z^2} \frac{\delta^2}{1 + 1/12\delta^2}, \tag{4}$$

where $\delta^2 u = u_{j-1} - 2u_j + u_{j+1}$ is a second difference operator.

The advantage of this scheme is that, like the 2nd order scheme, it leads to a tridiagonal system of linear equations, written as follows

$$\left(\tau\left(1+\alpha\delta^2\right) + b_l\delta^2 + b_l\tau h_j\right) v_j + b_l\tau\alpha\delta^2\left(h_j v_j\right) = \left(\tau\left(1+\alpha\delta^2\right) + a_l\delta^2 + a_l\tau h_j\right) u_j + a_l\tau\alpha\delta^2\left(h_j u_j\right), \tag{5}$$

where $\alpha = 0$ for 2nd order scheme or $\alpha = 1/12$ for 4th order scheme, $\tau = (k\Delta z)^2$, $h_j = m_j^2 - 1$.

For quantitative analysis of the numerical scheme, we introduce a discrete dispersion relation expressed in terms of a discrete horizontal wavenumber [31]

$$k_x^d(\Delta x, \Delta z, a_1 \dots a_p, b_1 \dots b_p, \theta) \;=\; k + \frac{\ln \prod_{l=1}^{p} t_l}{i\Delta x}, \tag{6}$$

$$t_l = \frac{1 - \frac{a_l}{k^2}\frac{4}{\Delta z^2}\left(\sin^2\left(\frac{k\sin\theta\Delta z}{2}\right) + 4\alpha\sin^4\left(\frac{k\sin\theta\Delta z}{2}\right)\right)}{1 - \frac{b_l}{k^2}\frac{4}{\Delta z^2}\left(\sin^2\left(\frac{k\sin\theta\Delta z}{2}\right) + 4\alpha\sin^4\left(\frac{k\sin\theta\Delta z}{2}\right)\right)}.$$

The specified relation includes the parameters of the computational grid, coefficients of the Padé approximation, and propagation angle θ.

III. 4TH ORDER DISCRETE NLBC

A distinctive feature of the propagation over the Earth's surface is the linear growth of the function $m^2(x,z)$ with height caused by the Pekeris transformation [32], [2]. Operator L above the computational domain ($z > z_{max}$) takes the form

$$Lu^n = \frac{1}{k^2}\frac{\partial^2}{\partial z^2}u^n + (\beta + \gamma z)\, u^n,$$

where β and γ are known constants defined from the function $m^2(x,z)$. Applying $\mathcal{Z}$-transform to the system (3), we get the following matrix system of equations [24]

$$\frac{\partial^2}{\partial z^2}\hat{\boldsymbol{\chi}} = k^2\left(\mathbf{J} - (\beta + \gamma z\mathbf{I})\right)\hat{\boldsymbol{\chi}}, \tag{7}$$

$$\hat{\boldsymbol{u}}(z) = \mathbf{P}\hat{\boldsymbol{\chi}}(z),$$

$$\hat{\boldsymbol{u}}(z) = \begin{pmatrix} \hat{v}_1(z) & \dots & \hat{v}_{p-1}(z) & \hat{u}(z) \end{pmatrix}^T,$$

$$\hat{u}(\xi) = \mathcal{Z}\{u^n\} = \sum_{n=0}^{\infty}\xi^{-n}u^n,$$

matrices $\mathbf{P}$ and $\mathbf{J}$ are the solutions of the generalized eigenvalue problem

$$\mathbf{B}(\xi) = \mathbf{A}(\xi)\,\mathbf{P}(\xi)\,\mathbf{J}(\xi)\,\mathbf{P}^{-1}(\xi),$$

$$\mathbf{A} = \begin{bmatrix} \xi b_1 & & & -a_1 \\ -a_2 & b_2 & & \\ & \ddots & \ddots & \\ & & -a_p & b_p \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} -\xi & & & 1 \\ 1 & -1 & & \\ & \ddots & \ddots & \\ & & 1 & -1 \end{bmatrix},$$

$$\mathbf{J} = diag\{s_1, s_2 \dots s_p\}.$$

The system of equations (7) consists of p independent differential equations of the form

$$\frac{\partial^2}{\partial z^2}\chi_l = k^2\left(s_l - \beta - \gamma z\right)\chi_l.$$

Replacing the second derivative with the 4th order FD approximation (4), we get to the following difference relation

$$(a_1 + bj)\chi_{j+1} + (c + dj)\chi_j + (a_{-1} + bj)\chi_{j-1} = 0,$$

$$b = \alpha\gamma\Delta z\tau,\; d = \gamma\Delta z\tau - 2b,$$

$$a_{\pm 1} = 1 - \alpha\tau\left(s_l - \beta\right) \pm b,$$

$$c = -2 + 2\alpha\tau\left(s_l - \beta\right) - \tau\left(s_l - \beta\right).$$

One can write the following expression for vector $\hat{\boldsymbol{u}}_j(\xi)$

$$\hat{\boldsymbol{u}}_j(\xi) = \mathbf{P}diag\{\chi_{j,1}, \chi_{j,2}, \dots, \chi_{j,p}\}\begin{pmatrix} 1 & 1 & \dots & 1 \end{pmatrix}^T.$$

Based on this expression, the condition on the upper boundary ($j = J$, $J\Delta z = z_{max}$) takes the form

$$\hat{\boldsymbol{u}}_{J-1}(\xi) = \hat{\mathbf{D}}(\xi)\hat{\boldsymbol{u}}_J(\xi),$$

$$\hat{\mathbf{D}}(\xi) = \mathbf{P}diag\left\{\frac{\chi_{J-1,1}}{\chi_{J,1}}, \frac{\chi_{J-1,2}}{\chi_{J,2}}, \cdots, \frac{\chi_{J-1,p}}{\chi_{J,p}}\right\}\mathbf{P}^{-1}.$$

Then the discrete NLBC can be written as follows [24]

$$\boldsymbol{u}_{J-1} - \mathbf{D}^0\boldsymbol{u}_J^n = \sum_{m=1}^{n-1}\mathbf{D}^m\boldsymbol{u}_J^{n-m},$$

$$\mathbf{D}^m = \frac{\tau^m}{2\pi}\int_0^{2\pi}\hat{\mathbf{D}}\left(\tau e^{i\varphi}\right)e^{im\varphi}d\varphi,\; \tau > 1. \tag{8}$$

An adaptive version of the Filon-Clenshaw-Curtis quadrature [33] is used for the numerical calculation of the integral (8).

Next, we need a constructive method for calculating the ratio $\chi_{l,J-1}/\chi_{l,J}$ for any value of ξ. Rewrite mentioned ratio as follows

$$\frac{\chi_{j-1}}{\chi_j} = -\frac{c + dj}{a_{-1} + bj} - \frac{a_1 + bj}{a_{-1} + bj}\frac{\chi_{j+1}}{\chi_j}.$$

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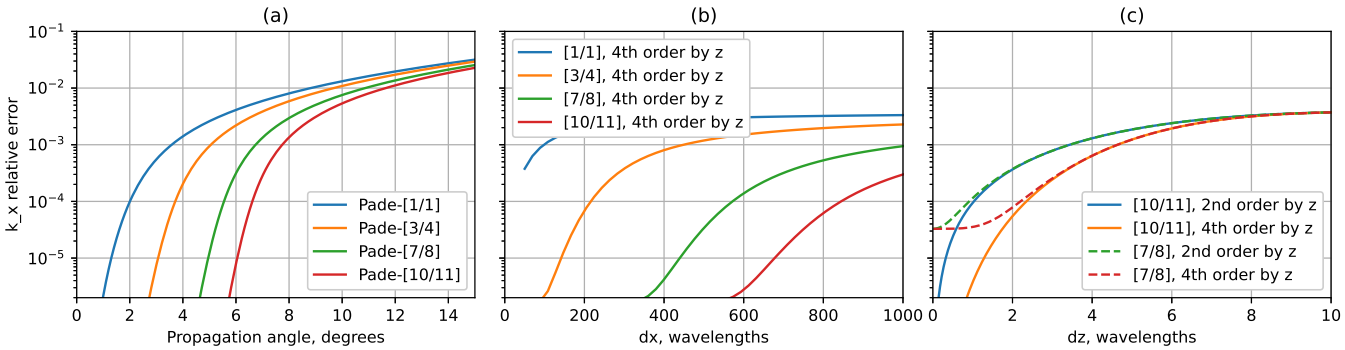


Fig. 1. The dependence of the relative error of discrete horizontal wavenumber k_x^d on the computational parameters. (a) k_x^d vs propagation angle θ, $\Delta x = 500\lambda$, $\Delta z \to 0$. (b) k_x^d vs range step Δx, $\Delta z = 0.8\lambda$, $\theta_{max} = 5^\circ$. (c) k_x^d vs altitude step Δz, $\Delta x = 500\lambda$, $\theta_{max} = 5^\circ$.

Denoting

$$f_j = -\frac{c + dj}{a_{-1} + bj},\ g_j = -\frac{a_1 + bj}{a_{-1} + bj},$$

the ratio may be presented as a generalized continued fraction

$$\frac{\chi_{j-1}}{\chi_j} = f_j + g_j \frac{\chi_{j+1}}{\chi_j} = f_j + \frac{g_j}{f_{j+1} + \frac{g_{j+1}}{f_{j+2}+\ldots}}.$$

This continued fraction can be efficiently calculated using the Lentz algorithm [34].

It should be noted that the Numerov extension of transparent boundary conditions for the constant refractive index has been considered in [35], [28].

IV. Results and Error Analysis

Discrete dispersion relation (6) allows quantifying the accuracy of various FD approximations depending on the computational grid, propagation angle θ, and the approximation order. Fig. 1(a) depicts the dependence of the relative error of discrete horizontal wavenumber k_x^d on the propagation angle. It is clearly seen that the accuracy of the numerical scheme decreases with increasing propagation angle. However, the accuracy increases with the approximation order of the propagation operator. Next, we consider the dependence of the relative error on the value of the longitudinal step Δx, which is shown in Fig. 1(b). As expected, the accuracy decreases as the step Δx increases. Fig. 1(c) shows a comparison of 2nd and 4th order schemes by the transverse coordinate. One can see that the 4th order scheme gives an opportunity to achieve better accuracy on the same transversal step, than the 2nd order one. It is interesting to note that the propagator approximation order also affects the dependence of error on Δz. It is clear from Fig. 1 that increasing the order of approximation makes it possible to use a less dense grid, which positively affects the computation time.

Note that maximum propagation angle θ_{max} depends not only on Padé approximation order but also on grid steps Δx and Δz and the approximation order by z. The analysis of the discrete dispersion relation helps to trace this dependence. The proposed scheme has no theoretical restrictions on the maximum propagation angle, provided that the appropriate computational parameters are selected.

Discrete dispersion relation (6) is further used to select the optimal parameters of the numerical scheme. The optimal Padé approximation order and grid steps Δx and Δz are selected in such a way that the relative error of the discrete wavenumber as a function of propagation angle θ does not exceed a specified value on segment $[0, \theta_{max}]$. Among all the appropriate values, the ones that lead to the most sparse grid, and therefore the minimum calculation time, is selected.

As an example, we first consider propagation over irregular terrain, which profile is depicted in Fig. 2. The maximum slope of this profile is equal to 5°. Then, using geometric considerations, the maximum propagation angle θ_{max} also should not significantly exceed 5°. First, we consider propagation in a standard atmosphere without waveguide effects. A Gauss antenna [2] with a beam width of 4° is located at a height of 70 m and emits a signal at a frequency of 3 GHz. The two-dimensional distributions of the field amplitude calculated by the proposed method and the SSF method are shown in Fig. 2. The computational parameters are shown in Table I. The optimal sizes of the grid cells for FD schemes were selected using the analysis of dispersion relations. The parameters were selected from the condition that the relative error at the distance of 150 km from the source should not exceed 5E-3. The implementation of SSF from PETOOL [36] program was used. Both methods use staircase approximation of the terrain profile with an impedance boundary condition. It is clearly seen that using the proposed 4th order scheme allows one to use a more sparse grid by the transversal coordinate, which results in more than three times faster computation than the 2nd order scheme.

Fig. 2 (right) demonstrates false reflections from the upper boundary of the computational domain that occur when using the SSF method with the absorption layer. It can be seen from Fig. 3 (left) that this leads to an overestimation of the field and incorrect simulation results. It should be noted that increasing the size of the computational domain and the absorbing layer reduces the false reflections. However, owing to data in Table I, this simultaneously increases the time of computation. Note that there is currently no reliable algorithm for selecting the optimal height of the computational domain

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TABLE I
COMPUTATIONAL PARAMETERS OF THE METHODS UNDER CONSIDERATION. $\theta_{max} = 5°$.

Method	z_{max}, m	Δx, λ	Δz, λ	CPU time, s
Padé-[10/11], 4th order by z (proposed method)	300	500	0.8	22
Padé-[10/11], 2nd order by z	300	500	0.1	75
Padé-[1/1], 2nd order by z (Crank-Nicolson)	300	4	0.01	6818
SSF (false reflections)	300	500	3	5
SSF (extended height)	3000	500	3	17

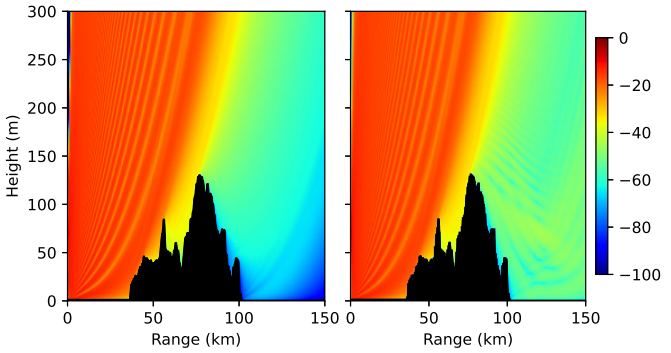


Fig. 2. Field component $10 \lg |u(x, z)|$. Propagation over irregular terrain in a standard atmosphere. Computation using the proposed method (Padé-[10/11]) (left) and SSF method (PETOOL) (right).

and the absorbing layer for the SSF method. At the same time, it is clearly seen that the constructed discrete NLBC allows avoiding undesirable false reflections from the upper boundary. The usual Crank-Nicolson scheme [14] requires too dense grid and cannot compete with other methods.

Fig. 3 (right) shows the application of the proposed method to propagation in an elevated duct [2]. The computational parameters are the same as in the example with the standard atmosphere.

In the last example, we compare 2nd and 4th order FD approximations. Fig. 4 (left) demonstrates spatial distribution of the difference between 2nd order scheme with $\Delta z = 0.1\lambda$ and 4th order scheme with $\Delta z = 0.8\lambda$. One can see that the difference between the results is insignificant. Fig. 4 (right) demonstrates the analogous comparison when calculations have been performed on the same grid ($\Delta z = 0.8\lambda$). It is observable that the error in some regions exceeds 5 dB and the error grows with the growth of the propagation angle. Thus, the use of the Numerov scheme is justified.

The stability of FD Padé approximations in solving differential equations has been considered in several studies [37], [2]. This method is widely used for calculating an acoustic field in underwater waveguides and its stability has been repeatedly proved theoretically and confirmed numerically [27], [12]. The Numerov method is also known as stable [38]. Padé approximations of the exponential operator of the form $[n - 1/n]$ and $[n - 2/n]$ are L-stable [37] and recommended to use in electromagnetic scattering problems [24], [2]. These works, as well as the above numerical experiments, suggest

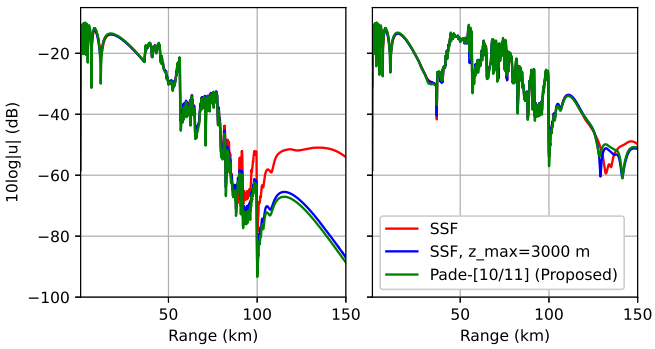


Fig. 3. Propagation over irregular terrain in a standard atmosphere (left) and elevated duct (right). Field amplitude at a height of 10 m above the surface.

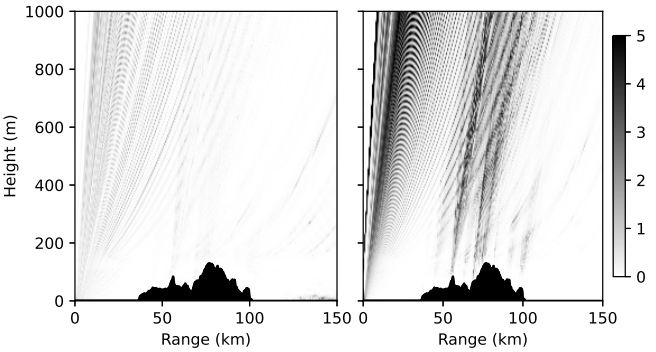


Fig. 4. Absolute error ($|10 \lg |u_{2nd}(x, z)| - 10 \lg |u_{4th}(x, z)||$) between 2nd and 4th order schemes (Padé-[10/11], $\Delta x = 500\lambda$). (left) 4th-order ($\Delta z = 0.8\lambda$) vs 2nd-order ($\Delta z = 0.1\lambda$). (right) 4th-order ($\Delta z = 0.8\lambda$) vs 2nd-order ($\Delta z = 0.8\lambda$).

that the proposed combination of Padé approximations and the Numerov method is also absolutely stable.

V. CONCLUSION

The SSF method remains the most efficient in terms of computation speed. However, the use of higher-order FD schemes allows approaching the efficiency of the SSF method. At the same time, the FD methods allow for the correct modeling of BC. In addition, FD methods are subject to quantitative error analysis, which contributes to the reliability of the obtained results and allows for selecting computational parameters depending on the propagation conditions. It is expected that using even higher-order FD schemes can further improve performance.

The proposed method is implemented in an open-source Python 3 library [39].

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